Programming on Modern Processors: the Single Thread Version

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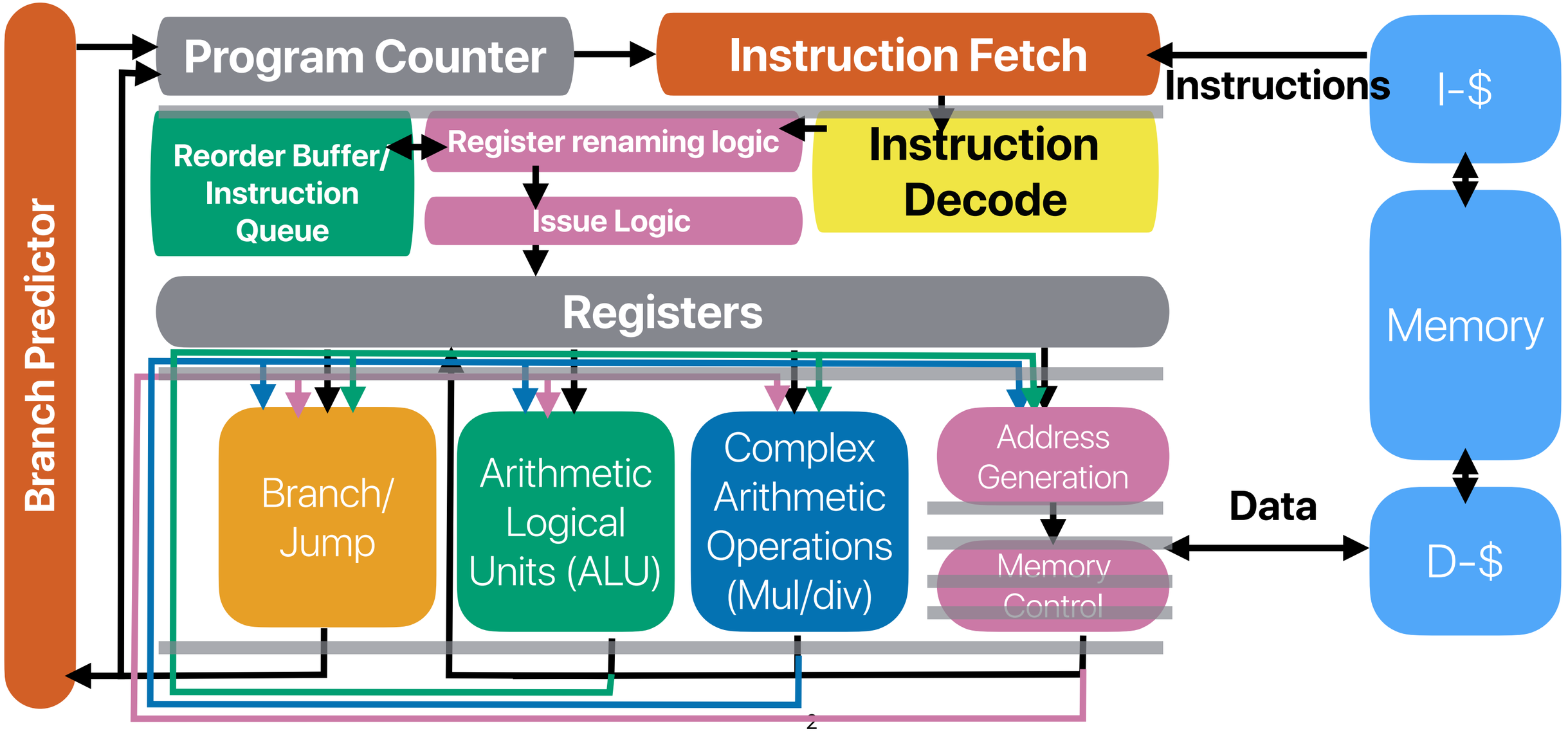
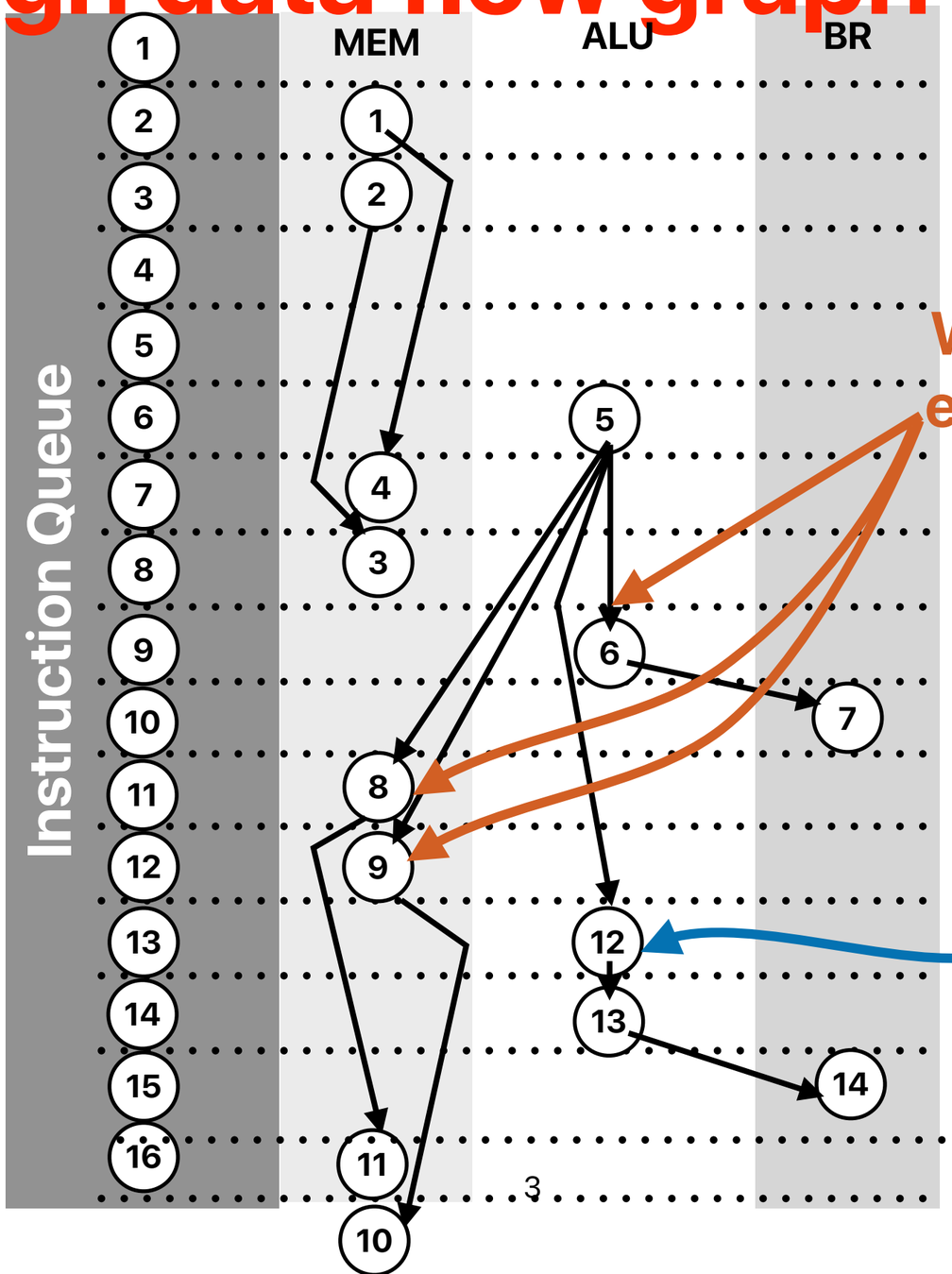
Recap: Register renaming + OoO + RoB



Through data flow graph analysis

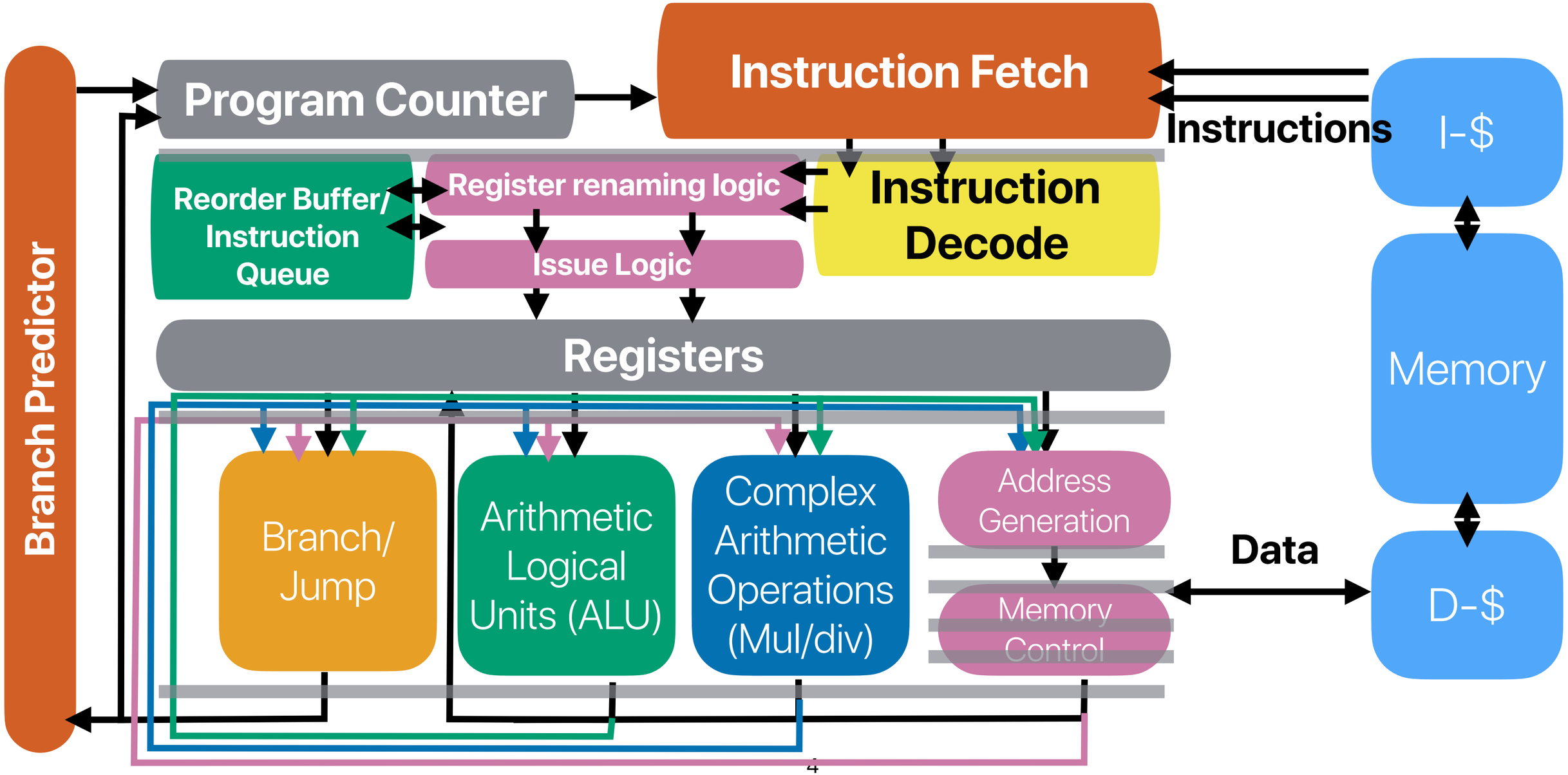
```
①  movq (%rdi,%rax), %rsi
②  movq (%rcx,%rax), %r8
③  movq %r8, (%rdi,%rax)
④  movq %rsi, (%rcx,%rax)
⑤  addq $8, %rax
⑥  cmpq %r9, %rax
⑦  jne  .L9
⑧  movq (%rdi,%rax), %rsi
⑨  movq (%rcx,%rax), %r8
⑩  movq %r8, (%rdi,%rax)
⑪  movq %rsi, (%rcx,%rax)
⑫  addq $8, %rax
⑬  cmpq %r9, %rax
⑭  jne  .L9
⑮  movq (%rdi,%rax), %rsi
⑯  movq (%rcx,%rax), %r8
⑰  movq %r8, (%rdi,%rax)
⑱  movq %rsi, (%rcx,%rax)
⑲  addq $8, %rax
⑳  cmpq %r9, %rax
㉑  jne  .L9
```



We cannot issue them earlier simply because structural hazards!

We could have this executed earlier if it's in the queue earlier

Recap: Register renaming + OoO + ROB + SuperScalar



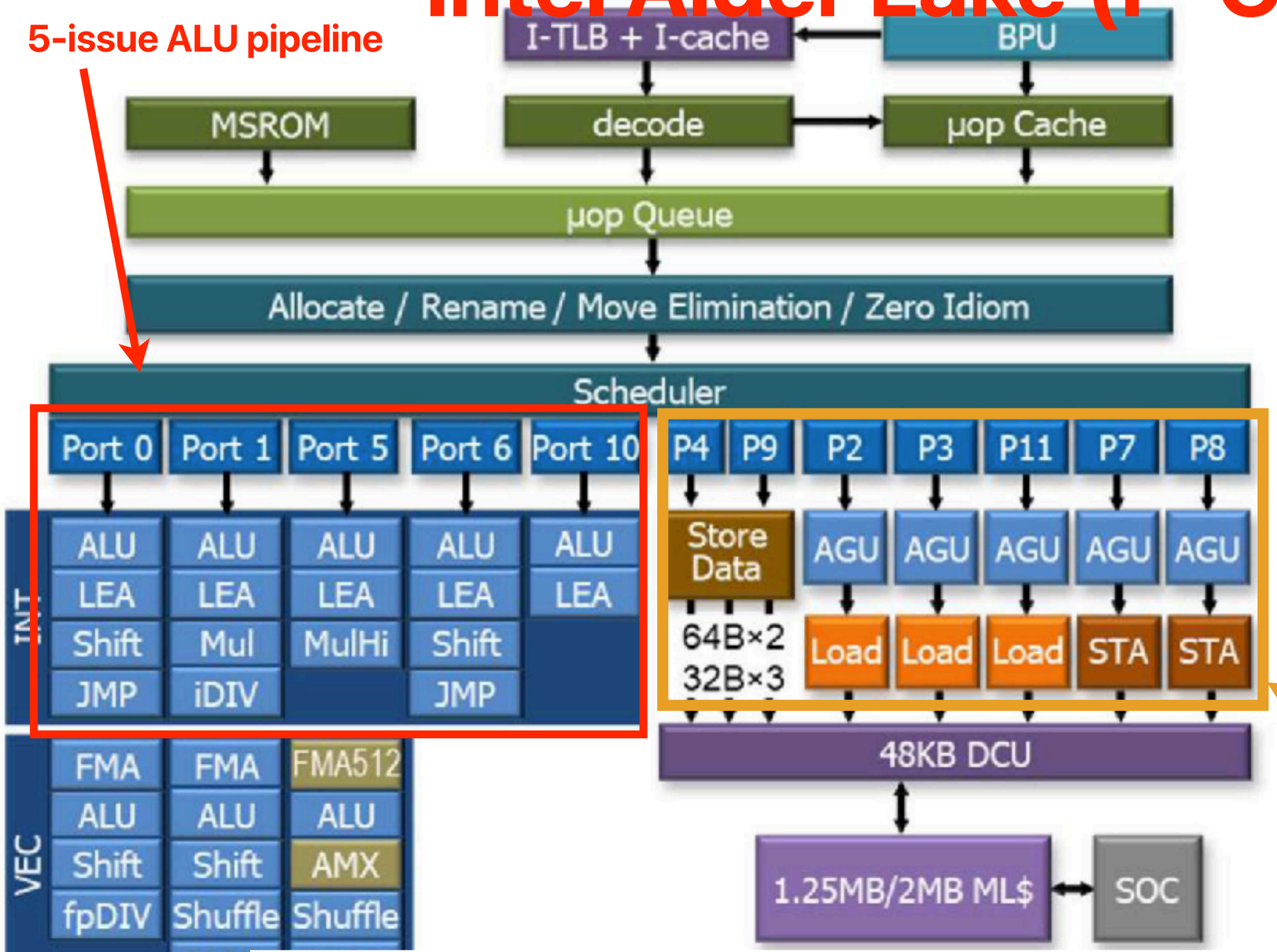
Intel Alder Lake (P-Core)

$$MinCPI = \frac{1}{12}$$

$$MinINTInst.CPI = \frac{1}{5}$$

$$MinMEMInst.CPI = \frac{1}{7}$$

$$MinBRInst.CPI = \frac{1}{2}$$



https://download.intel.com/newsroom/2021/client-computing/intel-architecture-day-2021-presentation.pdf

Recap: Super-Scalar + Register Renaming + Speculative Execution

- SuperScalar: fetching & issuing multiple instructions from the same process/thread/running program at the same cycle
- Register Renaming & OoO Scheduling
  - Redirecting the output of an instruction instance to a physical register
  - Redirecting inputs of an instruction instance from architectural registers to correct physical registers
  - Executing an instruction all operands are ready (the values of depending physical registers are generated)
- Speculative execution: execute an instruction before the processor know if we need to execute or not
  - Storing results in **reorder buffer** before the processor knows if the instruction is going to be executed or not.
  - Retiring instructions only when all earlier-order instructions are retired

Summary: Characteristics of modern processor architectures

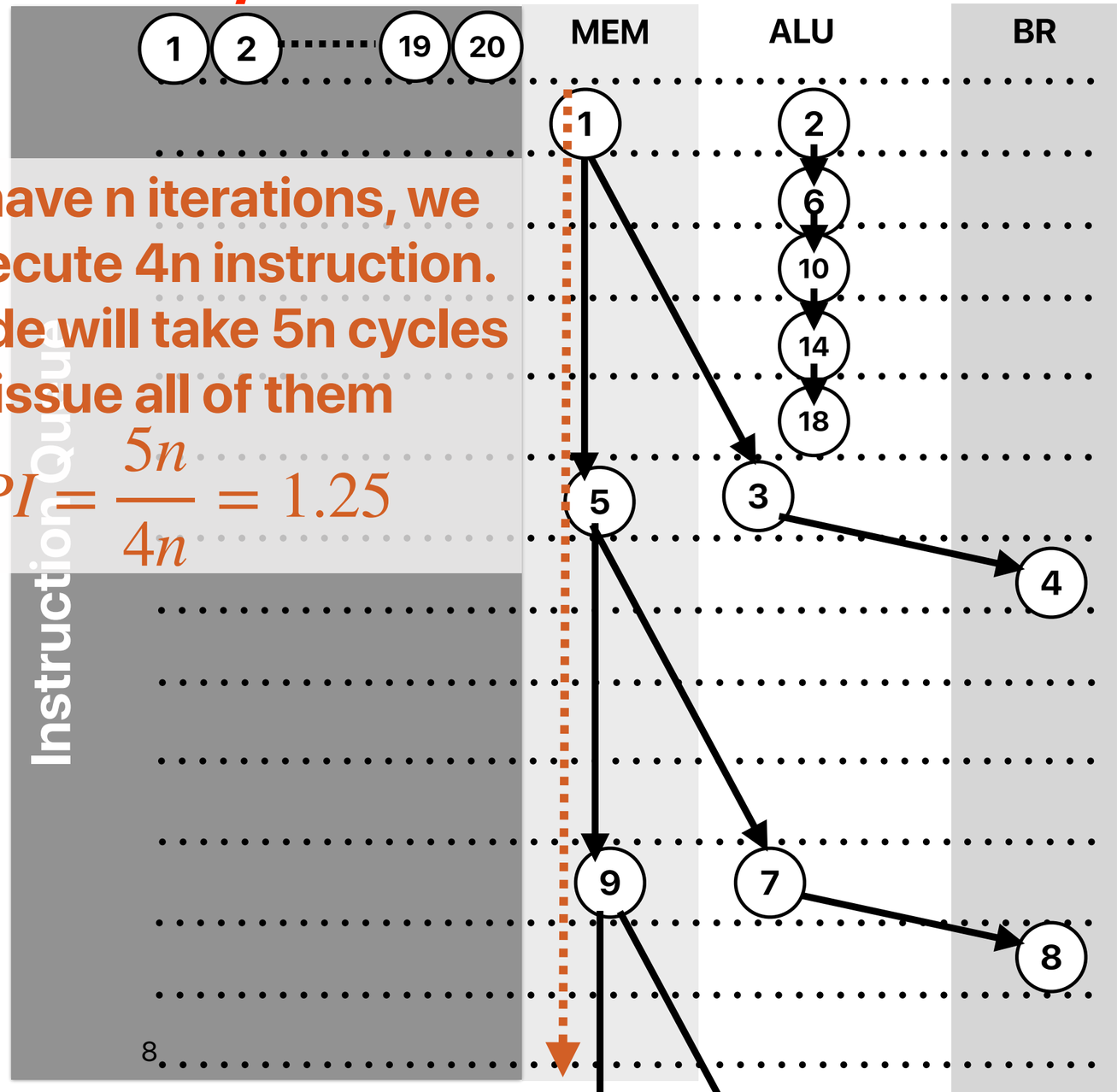
- Multiple-issue pipelines with multiple functional units available
  - Multiple ALUs
  - Multiple Load/store units
  - Dynamic OoO scheduling to reorder instructions whenever possible
- Cache — very high hit rate if your code has good locality
  - Very matured data/instruction prefetcher
- Branch predictors — very high accuracy if your code is predictable
  - Perceptron
  - Tournament predictors

What if we have “unlimited” fetch/issue width — “linked list”

```
① .L3:    movq   8(%rdi), %rdi
②         addl   $1, %eax
③         testq  %rdi, %rdi
④         jne    .L3
⑤ .L3:    movq   8(%rdi), %rdi
⑥         addl   $1, %eax
⑦         testq  %rdi, %rdi
⑧         jne    .L3
⑨ .L3:    movq   8(%rdi), %rdi
⑩         addl   $1, %eax
⑪         testq  %rdi, %rdi
⑫         jne    .L3
⑬ .L3:    movq   8(%rdi), %rdi
⑭         addl   $1, %eax
⑮         testq  %rdi, %rdi
⑯         jne    .L3
⑰ .L3:    movq   8(%rdi), %rdi
⑱         addl   $1, %eax
⑲         testq  %rdi, %rdi
⑳         jne    .L3
```

If we have n iterations, we will execute 4n instruction. The code will take 5n cycles to issue all of them

$$CPI = \frac{5n}{4n} = 1.25$$

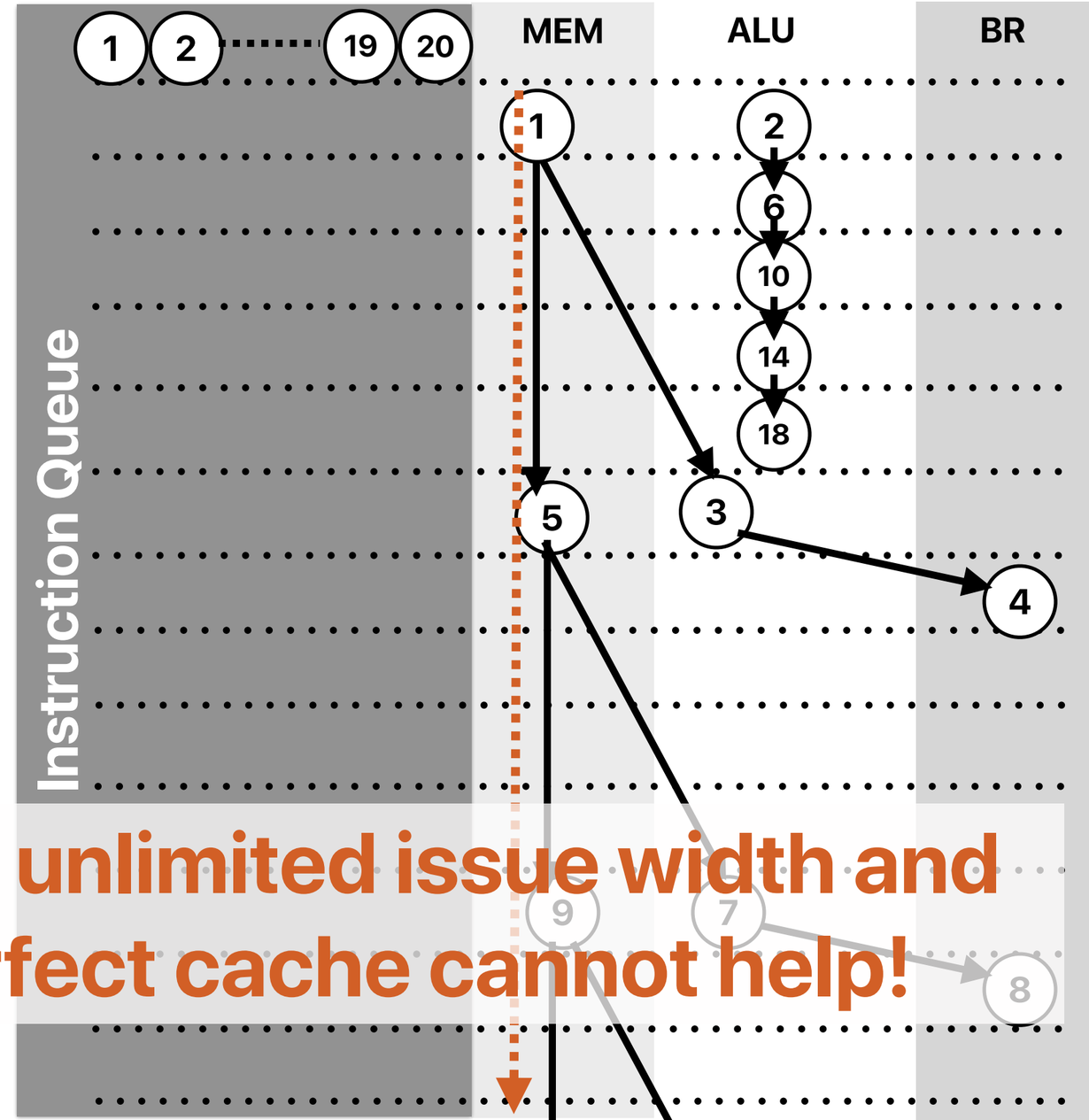


What if we have "unlimited" fetch/issue width — "linked list"

If we cannot improve the performance of executing
`movq 8(%rdi), %rdi`
we cannot improve the execution time.
That's the "critical path"!

```
do {
    number_of_nodes++;
    current = current->next;
} while ( current != NULL );
```

```
① .L3:   movq   8(%rdi), %rdi
②        addl   $1, %eax
③        testq  %rdi, %rdi
④        jne    .L3
```



Even unlimited issue width and a perfect cache cannot help!

Takeaways: programming modern processors

- The key to efficient code is exploiting as much instruction-level parallelism (ILP) or say higher instructions per cycle (IPC) or lower cycles per instruction (CPI) as possible
  - Avoiding data dependent operations (e.g., pointer chasing)
  - Avoiding inter-iteration dependencies (e.g., linked list, trees)

Outline

- Programming on modern processors — exploiting instruction-level parallelism
- Simultaneous multithreading

Problem: Popcount

- The population count (or popcount) of a specific value is the number of set bits (i.e., bits in 1s) in that value.
- Applications
  - Parity bits in error correction/detection code
  - Cryptography
  - Sparse matrix
  - Molecular Fingerprinting
  - Implementation of some succinct data structures like bit vectors and wavelet trees.

Problem: Popcount

- Given a 64-bit integer number, find the number of 1s in its binary representation.
- Example 1:

```
Input:  59487
Output: 9
```

Explanation: 59487's binary representation is `0b1011001010001111`

```
int main(int argc, char *argv[]) {

     uint64_t key = 0xdeadbeef;

     int count = 1000000000;
     uint64_t sum = 0;

     for (int i=0; i < count; i++)
     {
          sum += popcount(RandLFSR(key));
     }
     printf("Result: %lu\n", sum);
     return sum;
}
```

Five implementations

- Which of the following implementations will perform the best on modern pipeline processors?

A

```
inline int popcount(uint64_t x){
  int c=0;
  while(x)  {
        c += x & 1;
        x = x >> 1;
    }
    return c;
}
```

B

```
inline int popcount(uint64_t x) {
   int c = 0;
   while(x)     {
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
   }
   return c;
}
```

C

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     while(x)     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
}
```

D

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     for (uint64_t i = 0; i < 16; i++)
     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
  }
```

E

```
inline int popcount(uint64_t x) {
     int c = 0;
     for (uint64_t i = 0; i < 16; i++)
     {
         switch((x & 0xF))
         {
             case 1: c+=1; break;
             case 2: c+=1; break;
             case 3: c+=2; break;
             case 4: c+=1; break;
             case 5: c+=2; break;
             case 6: c+=2; break;
             case 7: c+=3; break;
             case 8: c+=1; break;
             case 9: c+=2; break;
             case 10: c+=2; break;
             case 11: c+=3; break;
             case 12: c+=2; break;
             case 13: c+=3; break;
             case 14: c+=3; break;
             case 15: c+=4; break;
             default: break;
         }
         x = x >> 4;
     }
     return c;
}
```

Five implementations

- Which of the following implementations will perform the best on modern pipeline processors?

A

```
inline int popcount(uint64_t x){
  int c=0;
  while(x)  {
        c += x & 1;
        x = x >> 1;
    }
    return c;
}
```

B

```
inline int popcount(uint64_t x) {
   int c = 0;
   while(x)     {
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
   }
   return c;
}
```

C

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     while(x)     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
}
```

D

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     for (uint64_t i = 0; i < 16; i++)
     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
 }
```

E

```
inline int popcount(uint64_t x) {
     int c = 0;
     for (uint64_t i = 0; i < 16; i++)
     {
         switch((x & 0xF))
         {
             case 1: c+=1; break;
             case 2: c+=1; break;
             case 3: c+=2; break;
             case 4: c+=1; break;
             case 5: c+=2; break;
             case 6: c+=2; break;
             case 7: c+=3; break;
             case 8: c+=1; break;
             case 9: c+=2; break;
             case 10: c+=2; break;
             case 11: c+=3; break;
             case 12: c+=2; break;
             case 13: c+=3; break;
             case 14: c+=3; break;
             case 15: c+=4; break;
             default: break;
         }
         x = x >> 4;
     }
     return c;
}
```

Why is B better than A?

- How many of the following statements explains the reason why B outperforms A with compiler optimizations
  - ① B has lower dynamic instruction count than A
  - ② B has significantly lower branch mis-prediction rate than A
  - ③ B has significantly fewer branch instructions than A
  - ④ B has better CPI than A

A. 0
B. 1
C. 2
D. 3
E. 4

A

```
inline int popcount(uint64_t x){
  int c=0;
  while(x)  {
        c += x & 1;
        x = x >> 1;
    }
    return c;
}
```

B

```
inline int popcount(uint64_t x) {
    int c = 0;
    while(x)      {
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
    }
    return c;
}
```

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A

```
inline int popcount(uint64_t x){
  int c=0;
  while(x)  {
        c += x & 1;
        x = x >> 1;
    }
    return c;
}
```

B

```
inline int popcount(uint64_t x) {
    int c = 0;
    while(x)      {
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
    }
    return c;
}
```

Why is B better than A?

A

```
inline int popcount(uint64_t x){
  int c=0;
  while(x)  {
        c += x & 1;
        x = x >> 1;
    }
    return c;
}
```

```
movl    %eax, %ecx
andl    $1, %ecx
addl    %ecx, %edx
shrq    %rax
jne     .L6
```

5*n instructions

B

```
inline int popcount(uint64_t x) {
   int c = 0;
   while(x)     {
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
     c += x & 1;
     x = x >> 1;
   }
   return c;
}
```

17*(n/4) = 4.25*n instructions

```
movl    %ecx, %eax
andl    $1, %eax
addl    %edx, %eax
movq    %rcx, %rdx
shrq    %rdx
andl    $1, %edx
addl    %eax, %edx
movq    %rcx, %rax
shrq    $2, %rax
andl    $1, %eax
addl    %edx, %eax
movq    %rcx, %rdx
shrq    $3, %rdx
andl    $1, %edx
addl    %eax, %edx
shrq    $4, %rcx
jne     .L6
```

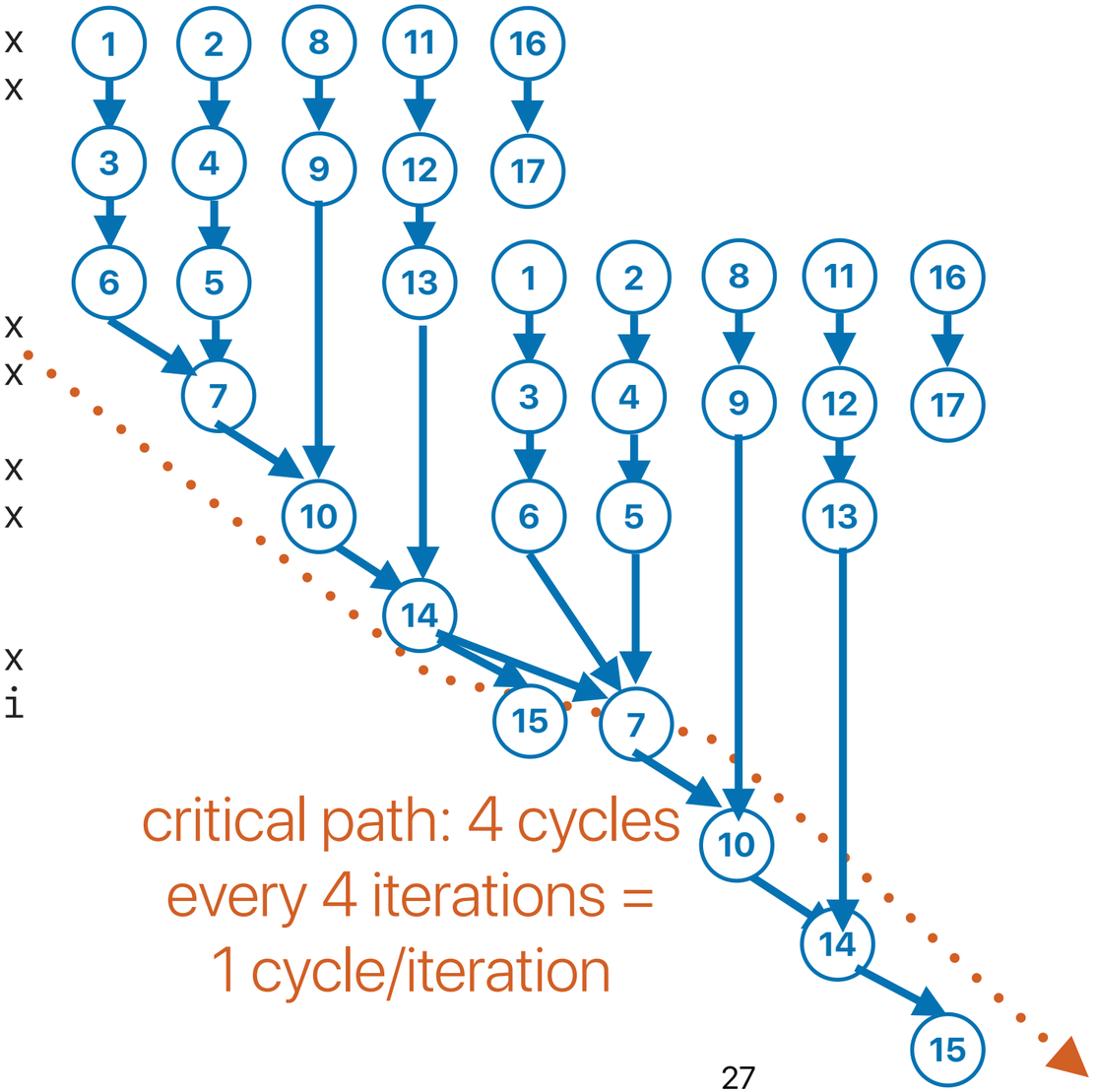
```
movq  %rdx, %rax
movq  %rdx, %rcx
shrq  %rax
shrq  $2, %rcx
andl  $1, %ecx
andl  $1, %eax
addl  %ecx, %eax
movl  %edx, %ecx
andl  $1, %ecx
addl  %ecx, %eax
movq  %rdx, %rcx
shrq  $3, %rcx
andl  $1, %ecx
addl  %ecx, %eax
addl  %eax, %esi
shrq  $4, %rdx
jne   .L6
```

Only one branch for four iterations in A

Why is B better than A?

B

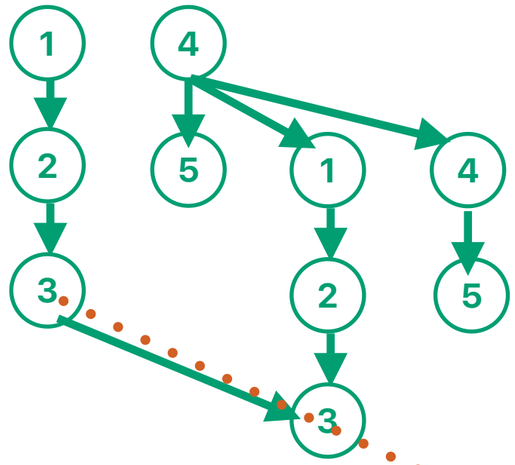
```
①  movq  %rdx, %rax
②  movq  %rdx, %rcx
③  shrq  %rax
④  shrq  $2, %rcx
⑤  andl  $1, %ecx
⑥  andl  $1, %eax
⑦  addl  %ecx, %eax
⑧  movl  %edx, %ecx
⑨  andl  $1, %ecx
⑩  addl  %ecx, %eax
⑪  movq  %rdx, %rcx
⑫  shrq  $3, %rcx
⑬  andl  $1, %ecx
⑭  addl  %ecx, %eax
⑮  addl  %eax, %esi
⑯  shrq  $4, %rdx
⑰  jne   .L6
```



critical path: 4 cycles
every 4 iterations =
1 cycle/iteration

A

```
①  movl  %eax, %ecx
②  andl  $1, %ecx
③  addl  %ecx, %edx
④  shrq  %rax
⑤  jne   .L6
```



critical path:
1 cycles/iteration

Why is B better than A?

- How many of the following statements explains the reason why B outperforms A with compiler optimizations
  - ① ✓ B has lower dynamic instruction count than A
  - ② B has significantly lower branch mis-prediction rate than A
  - ③ ✓ B has significantly fewer branch instructions than A
  - ④ ✓ B has better CPI

A. 0
B. 1
C. 2
D. 3
E. 4

A

```
inline int popcount(uint64_t x){
  int c=0;
  while(x)  {
        c += x & 1;
        x = x >> 1;
    }
    return c;
}
```

B

```
inline int popcount(uint64_t x) {
    int c = 0;
    while(x)      {
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
    }
    return c;
}
```

Takeaways: programming modern processors

- The key to efficient code is exploiting as much instruction-level parallelism (ILP) or say higher instructions per cycle (IPC) or lower cycles per instruction (CPI) as possible
- Loop unrolling is effective as control overhead is still significant despite we have branch predictors and OoO.

Why is C better than B?

- How many of the following statements explains the reason why B outperforms C with compiler optimizations
  - ① C has lower dynamic instruction count than B
  - ② C has significantly lower branch mis-prediction rate than B
  - ③ C has significantly fewer branch instructions than B
  - ④ C has better CPI than B

A. 0
B. 1
C. 2
D. 3
E. 4

C

```
inline int popcount(uint64_t x) {
      int c = 0;
      int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
      while(x)      {
          c += table[(x & 0xF)];
          x = x >> 4;
      }
      return c;
}
```

B

```
inline int popcount(uint64_t x) {
    int c = 0;
    while(x)      {
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
    }
    return c;
}
```

Why is C better than B?

- How many of the following statements explains the reason why B outperforms C with compiler optimizations
  - ① C has lower dynamic instruction count than B
  - ② C has significantly lower branch mis-prediction rate than B
  - ③ C has significantly fewer branch instructions than B
  - ④ C has better CPI than B

  A. 0
  B. 1
  C. 2
  D. 3
  E. 4

C

```
inline int popcount(uint64_t x) {
      int c = 0;
      int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
      while(x)      {
          c += table[(x & 0xF)];
          x = x >> 4;
      }
      return c;
}
```

B

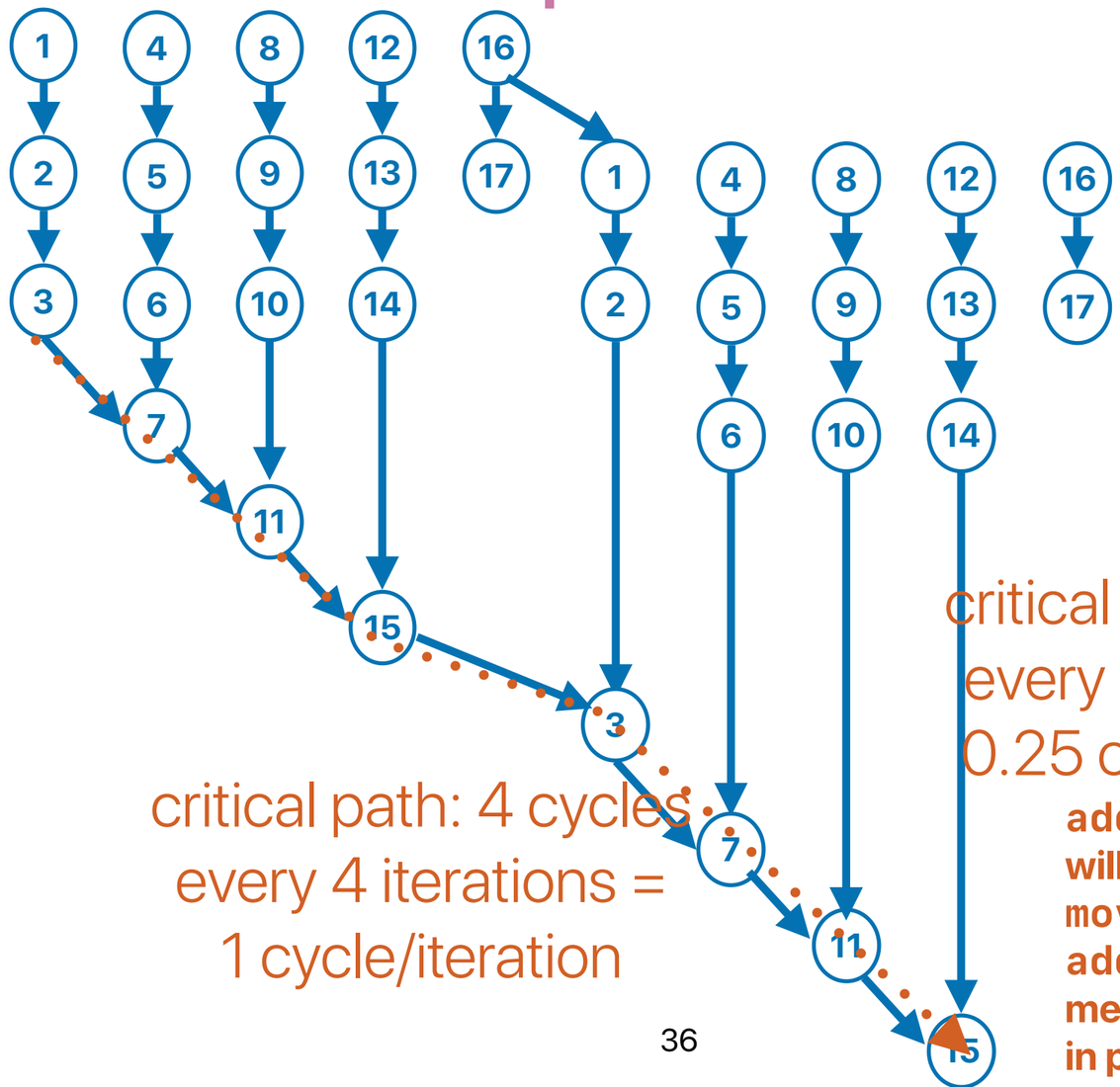
```
inline int popcount(uint64_t x) {
    int c = 0;
    while(x)      {
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
    }
    return c;
}
```

Why is C better than B?

B

```
① movl    %ecx, %eax
② andl    $1, %eax
③ addl    %edx, %eax
④ movq    %rcx, %rdx
⑤ shrq    %rdx
⑥ andl    $1, %edx
⑦ addl    %eax, %edx
⑧ movq    %rcx, %rax
⑨ shrq    $2, %rax
⑩ andl    $1, %eax
⑪ addl    %edx, %eax
⑫ movq    %rcx, %rdx
⑬ shrq    $3, %rdx
⑭ andl    $1, %edx
⑮ addl    %eax, %edx
⑯ shrq    $4, %rcx
⑰ jne     .L6
```

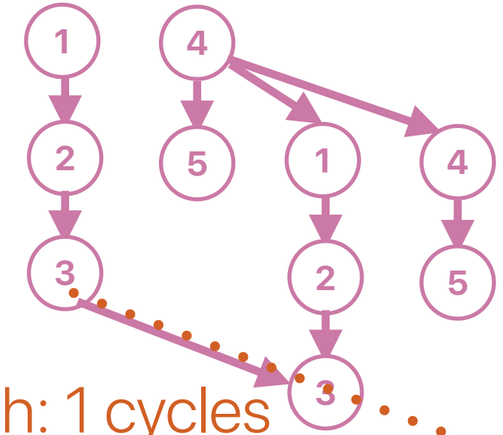
These 5 instructions represent 4 iterations!!!



critical path: 4 cycles
every 4 iterations =
1 cycle/iteration

C

```
.L6:
① movq  %rcx, %rdi
② andl  $15, %edi
③ addl  (%rsp,%rdi,4), %eax
④ shrq  $4, %rcx
⑤ jne   .L6
```



critical path: 1 cycles
every 4 iterations =
0.25 cycle/iteration

`addl (%rsp,%rdi,4), %eax`
will be translated into uOPs of
`mov (%rsp,%rdi,4), something` and
`addl something, %eax` –
memory operations can be executed in parallel

Why is C better than B?

- How many of the following statements explains the reason why B outperforms C with compiler optimizations
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  - ③ C has significantly fewer branch instructions than B
  - ④ C has better CPI than B

A. 0

B. 1

C. 2

D. 3

E. 4

— C only needs one load, one add, one shift, the same amount of iterat

— the same number being predicted.

— the same amount of branches

— Probably not. In fact, the load may have negative effect without architectural supports

C

```
inline int popcount(uint64_t x) {
      int c = 0;
      int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
      while(x)      {
          c += table[(x & 0xF)];
          x = x >> 4;
      }
      return c;
}
```

B

```
inline int popcount(uint64_t x) {
    int c = 0;
    while(x)      {
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
      c += x & 1;
      x = x >> 1;
    }
    return c;
}
```

Takeaways: programming modern processors

- The key to efficient code is exploiting as much instruction-level parallelism (ILP) or say higher instructions per cycle (IPC) or lower cycles per instruction (CPI) as possible
  - Avoiding data dependent operations (e.g., pointer chasing)
  - Avoiding inter-iteration dependencies (e.g., linked list, trees)
- Loop unrolling is effective as control overhead is still significant despite we have branch predictors and OoO.
- With caches, we can potentially use small lookup tables to replace more expensive data dependent operations

Why is D better than C?

- How many of the following statements explains the main reason why B outperforms C with compiler optimizations
  - ① D has lower dynamic instruction count than C
  - ② D has significantly lower branch mis-prediction rate than C
  - ③ D has significantly fewer branch instructions than C
  - ④ D has better CPI than C

A. 0
B. 1
C. 2
D. 3
E. 4

C

```
inline int popcount(uint64_t x) {
      int c = 0;
      int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
      while(x)      {
          c += table[(x & 0xF)];
          x = x >> 4;
      }
      return c;
}
```

D

```
inline int popcount(uint64_t x) {
      int c = 0;
      int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
      for (uint64_t i = 0; i < 16; i++)
      {
          c += table[(x & 0xF)];
          x = x >> 4;
      }
      return c;
 }
```

Why is D better than C?

- How many of the following statements explains the main reason why B outperforms C with compiler optimizations
  - ① D has lower dynamic instruction count than C
  - ② D has significantly lower branch mis-prediction rate than C
  - ③ D has significantly fewer branch instructions than C
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A. 0
B. 1
C. 2
D. 3
E. 4

C

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     while(x)     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
}
```

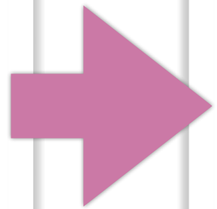
D

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     for (uint64_t i = 0; i < 16; i++)
     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
 }
```

Loop unrolling eliminates all branches!

D

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     for (uint64_t i = 0; i < 16; i++)
     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
 }
```



```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1, 2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
          c += table[(x & 0xF)];
          x = x >> 4;
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          x = x >> 4;
          c += table[(x & 0xF)];
          x = x >> 4;
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          x = x >> 4;
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          x = x >> 4;
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          c += table[(x & 0xF)];
          x = x >> 4;
          c += table[(x & 0xF)];
          x = x >> 4;
     return c;
}
```

Why is D better than C?

- How many of the following statements explains the main reason why B outperforms C with compiler optimizations
  - ① D has lower dynamic instruction count than C — Compiler can do loop unrolling — no branches
  - ② D has significantly lower branch mis-prediction rate than C — Could be
  - ③ D has significantly fewer branch instructions than C — maybe eliminated through loop unrolling…
  - ④ D has better CPI than C — about the same

A. 0
B. 1
C. 2
D. 3
E. 4

C

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     while(x)     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
}
```

D

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     for (uint64_t i = 0; i < 16; i++)
     {
         c += table[(x & 0xF)];
         x = x >> 4;
     }
     return c;
 }
```

Takeaways: programming modern processors

- The key to efficient code is exploiting as much instruction-level parallelism (ILP) or say higher instructions per cycle (IPC) or lower cycles per instruction (CPI) as possible
  - Avoiding data dependent operations (e.g., pointer chasing)
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- With caches, we can potentially use small lookup tables to replace more expensive data dependent operations
- Making your code more predictable is the key!
  - Compilers can confidently perform aggressive optimizations
  - Branch predictors can be more accurate
  - Cache miss rate can be really low

https://www.pollev.com/hungweitseng close in 1:30

Why is E the slowest?

- How many of the following statements explains the main reason why B outperforms C with compiler optimizations
  - ① E has the most dynamic instruction count
  - ② E has the highest branch mis-prediction rate
  - ③ E has the most branch instructions
  - ④ E can incur the most data hazards than others

A. 0
B. 1
C. 2
D. 3
E. 4

D

```
inline int popcount(uint64_t x) {
    int c = 0;
    int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
    for (uint64_t i = 0; i < 16; i++)
    {
        c += table[(x & 0xF)];
        x = x >> 4;
    }
    return c;
}
```

E

```
inline int popcount(uint64_t x) {
    int c = 0;
    for (uint64_t i = 0; i < 16; i++)
    {
        switch((x & 0xF))
        {
            case 1: c+=1; break;
            case 2: c+=1; break;
            case 3: c+=2; break;
            case 4: c+=1; break;
            case 5: c+=2; break;
            case 6: c+=2; break;
            case 7: c+=3; break;
            case 8: c+=1; break;
            case 9: c+=2; break;
            case 10: c+=2; break;
            case 11: c+=3; break;
            case 12: c+=2; break;
            case 13: c+=3; break;
            case 14: c+=3; break;
            case 15: c+=4; break;
            default: break;
        }
        x = x >> 4;
    }
    return c;
}
```

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A. 0
B. 1
C. 2
D. 3
E. 4

D

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     for (uint64_t i = 0; i < 16; i++)
     {
          c += table[(x & 0xF)];
          x = x >> 4;
     }
     return c;
 }
```

E

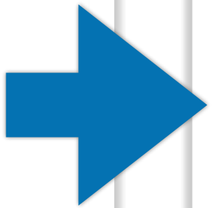
```
inline int popcount(uint64_t x) {
     int c = 0;
     for (uint64_t i = 0; i < 16; i++)
     {
          switch((x & 0xF))
          {
               case 1: c+=1; break;
               case 2: c+=1; break;
               case 3: c+=2; break;
               case 4: c+=1; break;
               case 5: c+=2; break;
               case 6: c+=2; break;
               case 7: c+=3; break;
               case 8: c+=1; break;
               case 9: c+=2; break;
               case 10: c+=2; break;
               case 11: c+=3; break;
               case 12: c+=2; break;
               case 13: c+=3; break;
               case 14: c+=3; break;
               case 15: c+=4; break;
               default: break;
          }
          x = x >> 4;
     }
     return c;
 }
```

Why is E the slowest?

It's not predicting "taken" or "not taken", it's about which address to jump — hard for BPUs

E

```
inline int popcount(uint64_t x) {
    int c = 0;
    for (uint64_t i = 0; i < 16; i++)
    {
        switch((x & 0xF))
        {
            case 1: c+=1; break;
            case 2: c+=1; break;
            case 3: c+=2; break;
            case 4: c+=1; break;
            case 5: c+=2; break;
            case 6: c+=2; break;
            case 7: c+=3; break;
            case 8: c+=1; break;
            case 9: c+=2; break;
            case 10: c+=2; break;
            case 11: c+=3; break;
            case 12: c+=2; break;
            case 13: c+=3; break;
            case 14: c+=3; break;
            case 15: c+=4; break;
            default: break;
        }
        x = x >> 4;
    }
    return c;
}
```



```
.L11:
        movq    %r9, %rcx
        andl    $15, %ecx
        movslq  (%r8,%rcx,4), %rcx
        addq    %r8, %rcx
        notrack jmp     *%rcx
.L7:
        .long   .L5-.L7
        .long   .L10-.L7
        .long   .L10-.L7
        .long   .L9-.L7
        .long   .L10-.L7
        .long   .L9-.L7
        .long   .L9-.L7
        .long   .L8-.L7
        .long   .L10-.L7
        .long   .L9-.L7
        .long   .L9-.L7
        .long   .L8-.L7
        .long   .L9-.L7
        .long   .L8-.L7
        .long   .L8-.L7
        .long   .L6-.L7
.L8:
        addl    $3, %eax
.L5:
        shrq    $4, %r9
        subq    $1, %rsi
        jne     .L11
        cltq
        addq    %rax, %rbx
        subl    $1, %edi
        jne     .L12
```

```
.L9:
        .cfi_restore_state
        addl    $2, %eax
        jmp     .L5
        .p2align 4,,10
        .p2align 3
.L10:
        addl    $1, %eax
        jmp     .L5
        .p2align 4,,10
        .p2align 3
.L6:
        addl    $4, %eax
        jmp     .L5
```

Why is E the slowest?

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A. 0
B. 1
C. 2
D. 3
E. 4

D

```
inline int popcount(uint64_t x) {
     int c = 0;
     int table[16] = {0, 1, 1, 2, 1,
2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4};
     for (uint64_t i = 0; i < 16; i++)
     {
          c += table[(x & 0xF)];
          x = x >> 4;
     }
     return c;
 }
```

E

```
inline int popcount(uint64_t x) {
     int c = 0;
     for (uint64_t i = 0; i < 16; i++)
     {
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          {
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               case 3: c+=2; break;
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               case 8: c+=1; break;
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               case 12: c+=2; break;
               case 13: c+=3; break;
               case 14: c+=3; break;
               case 15: c+=4; break;
               default: break;
          }
          x = x >> 4;
     }
     return c;
 }
```

Hardware acceleration

- Because popcount is important, both intel and AMD added a POPCNT instruction in their processors with SSE4.2 and SSE4a
- In C/C++, you may use the intrinsic "_mm_popcnt_u64" to get # of "1"s in an unsigned 64-bit number
  - You need to compile the program with -m64 -msse4.2 flags to enable these new features

```
#include <smmintrin.h>
inline int popcount(uint64_t x) {
      int c = _mm_popcnt_u64(x);
      return c;
}
```

Summary of popcounts

	Cycles	IC	IPC/ILP	ET	# of branches	Branch mis-prediction rate
A	334270949858	118508036769	2.821	23.237	65366730559	0.012
B	290179950840	68341242029	4.246	13.419	18319495534	0.004
C	102882218758	27580097715	3.730	5.380	18129328282	0.004
D	80043714833	19072124536	4.197	3.754	1037120144	0.000
E	253238690876	376641071475	0.672	73.977	73967642413	0.184
SSE4.2	22089754243	8444815558	2.616	1.654	1014543318	0.000

Takeaways: programming modern processors

- The key to efficient code is exploiting as much instruction-level parallelism (ILP) or say higher instructions per cycle (IPC) or lower cycles per instruction (CPI) as possible
  - Avoiding data dependent operations (e.g., pointer chasing)
  - Avoiding inter-iteration dependencies (e.g., linked list, trees)
- Loop unrolling is effective as control overhead is still significant despite we have branch predictors and OoO.
- With caches, we can potentially use small lookup tables to replace more expensive data dependent operations
- Making your code more predictable is the key!
  - Compilers can confidently perform aggressive optimizations
  - Branch predictors can be more accurate
  - Cache miss rate can be really low
- If there is a hardware feature supporting the desire computation — we should try it!

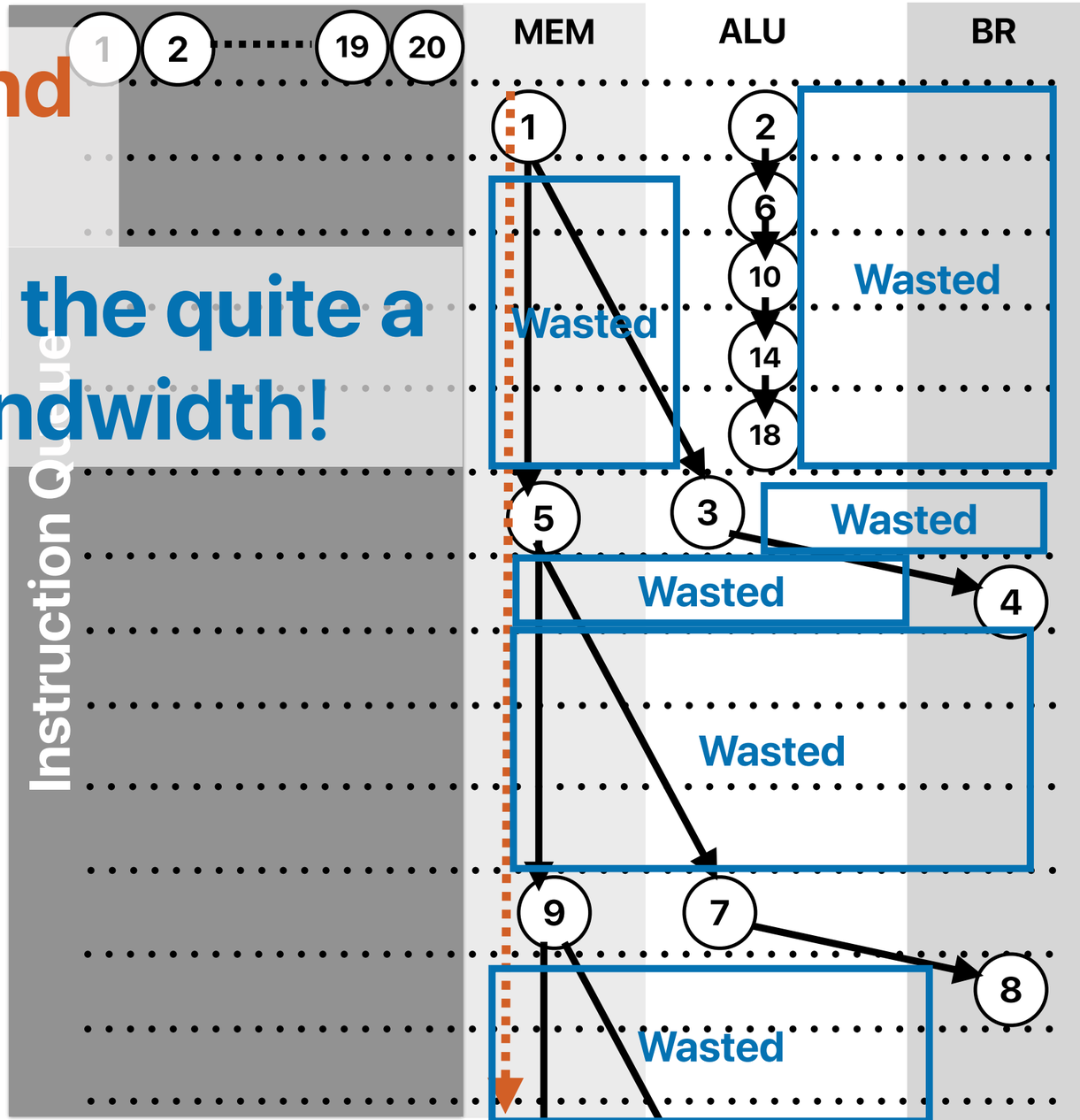
What if we have "unlimited" fetch/issue width — "linked list"

Even unlimited issue width and a perfect cache cannot help!

We're wasting the quite a lot issue bandwidth!

```
do {
    number_of_nodes++;
    current = current->next;
} while ( current != NULL );
```

```
① .L3:    movq    8(%rdi), %rdi
②         addl    $1, %eax
③         testq   %rdi, %rdi
④         jne     .L3
```



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Most of time, we can't fully utilize the function units

Top at 4.3

Best performing one at 2.7

Performance code may not even fully utilize FUs, either.

Outline

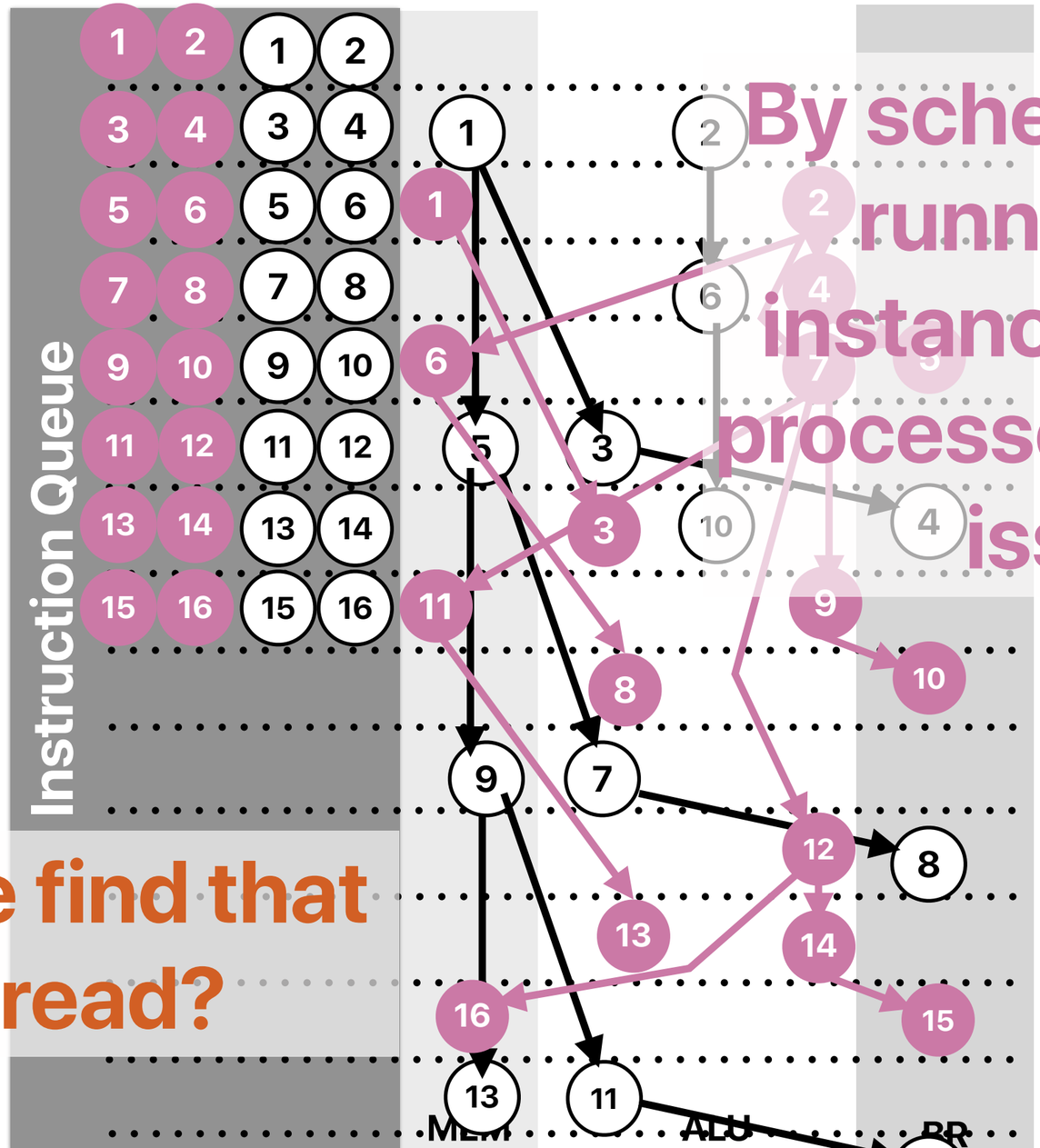
- Parallel architectures
  - Simultaneous multithreading (SMT)
  - Chip multiprocessors (CMP)
- Parallel programming

Parallel architectures

Simultaneous multithreading

Concept: Simultaneous Multithreading (SMT)

```
① movq    8(%rdi), %rdi
② addl    $1, %eax
③ testq   %rdi, %rdi
④ jne     .L3
⑤ movq    8(%rdi), %rdi
⑥ addl    $1, %eax
⑦ testq   %rdi, %rdi
⑧ jne     .L3
⑨ movq    8(%rdi), %rdi
⑩ addl    $1, %eax
⑪ testq   %rdi, %rdi
⑫ jne     .L3
```




```
① movl (%rdi), %ecx
② cmpq %rdx, %rdi
③ addl %ecx, %eax
④ jne  .L3
⑤ movl (%rdi), %ecx
⑥ addq $4, %rdi
⑦ addl %ecx, %eax
⑧ cmpq %rdx, %rdi
⑨ jne  .L3
⑩ movl (%rdi), %ecx
⑪ addq $4, %rdi
⑫ addl %ecx, %eax
⑬ cmpq %rdx, %rdi
⑭ jne  .L3
⑮ movl (%rdi), %ecx
⑯ addq $4, %rdi
```

By scheduling another running program instance (thread), the processor has 0 wasted issue slots!

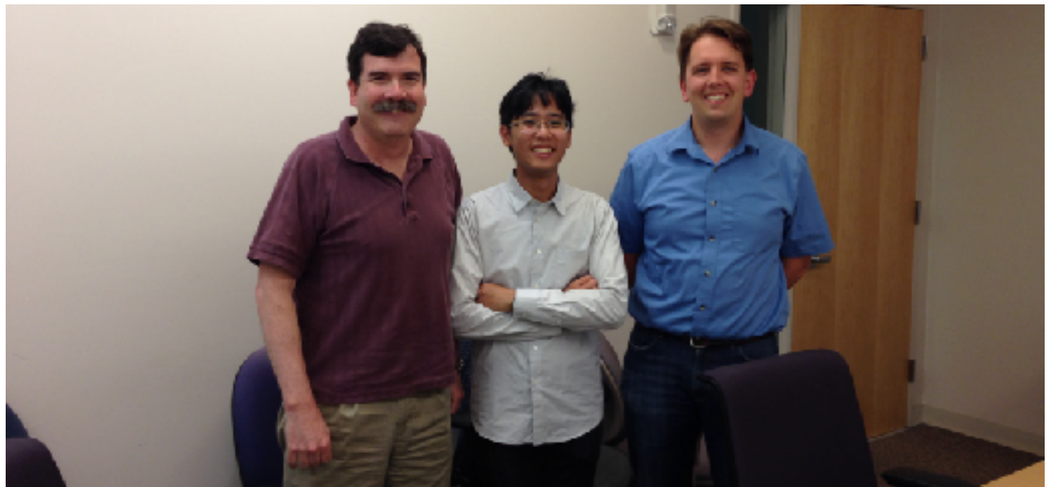
Where can we find that "other" thread?

Where to find another “thread”

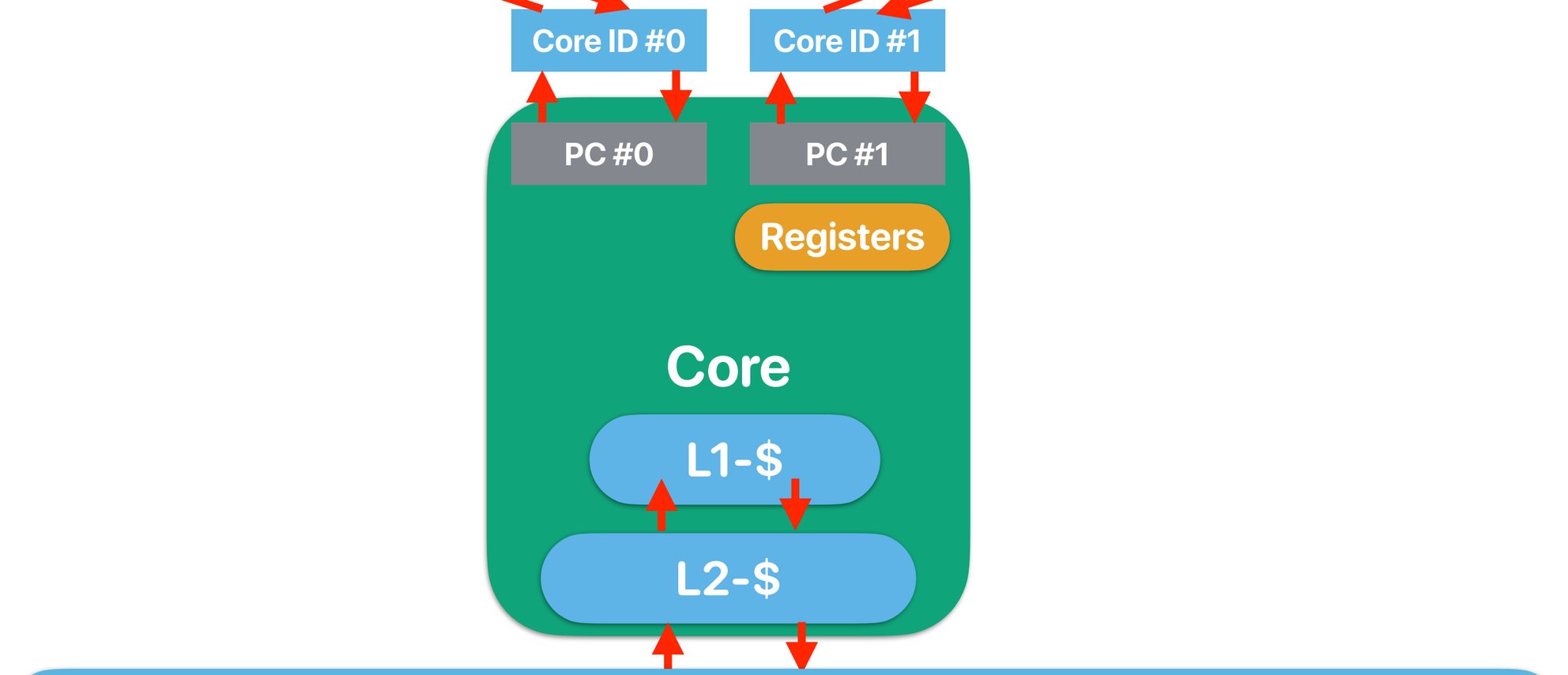
- Another process/running program forked from a completely different program
- Another process forked/cloned from the current process
  - The forked program cloned the memory content at the forked time
  - The forked program cannot view the memory space of the original process
  - The forked program can perform a subset of tasks form the original process
  - The forked and the original process can exchange data through files or inter-process communication APIs
- A software thread spawned from the current process
  - The spawned thread executes a function instance from the main process to offload computation from the main process
  - The spawned thread shares the memory space with the main process

Simultaneous multithreading

- The processor can schedule instructions from different threads/processes/ programs
- Fetch instructions from different threads/processes to fill the not utilized part of pipeline
  - Exploit “thread level parallelism” (TLP) to solve the problem of insufficient ILP in a single thread
  - You need to create an illusion of multiple processors for OSs
- Invented by Dean Tullsen (Now a professor at **UCSD CSE**)



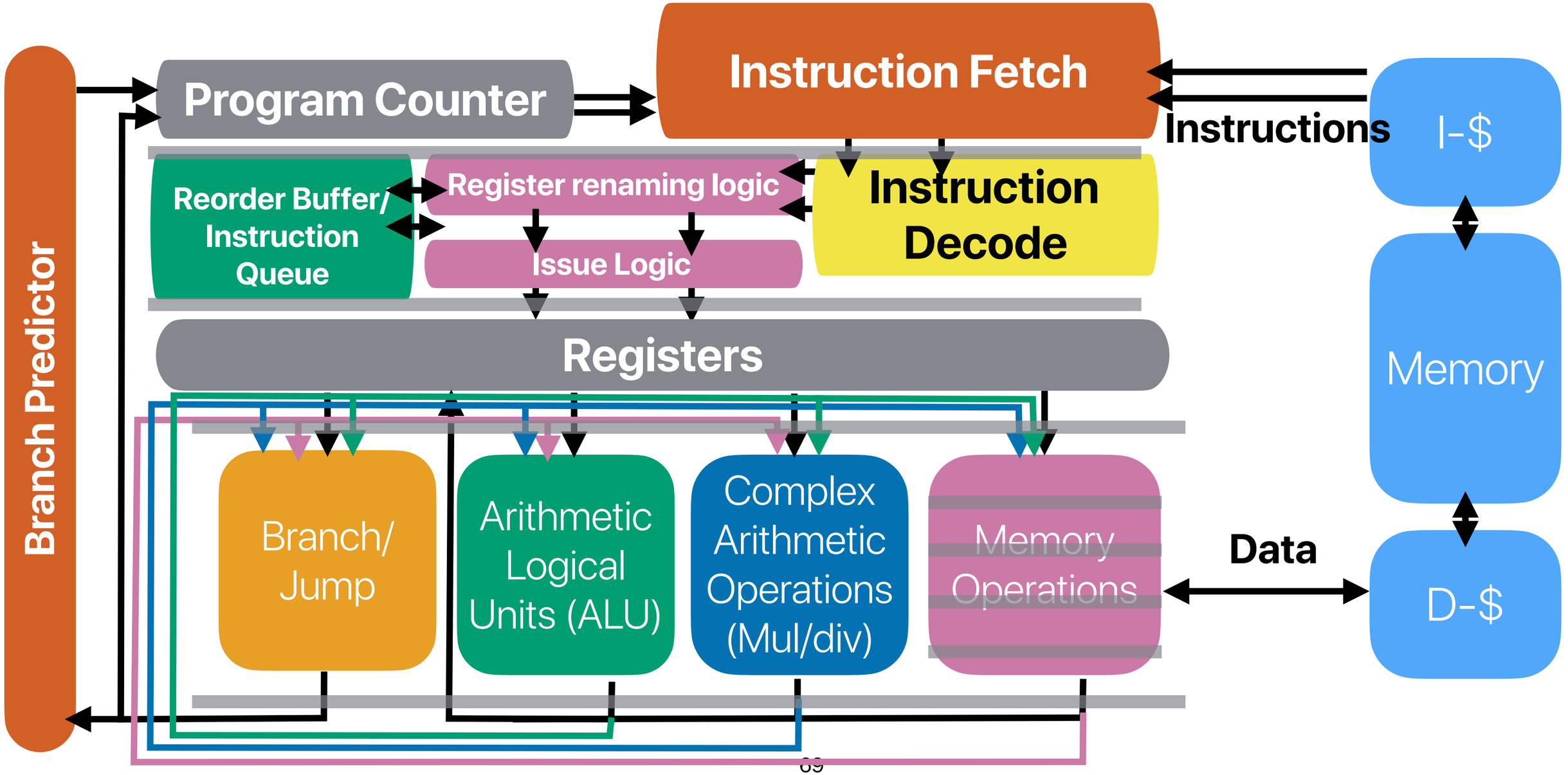
SMT from the user/OS' perspective



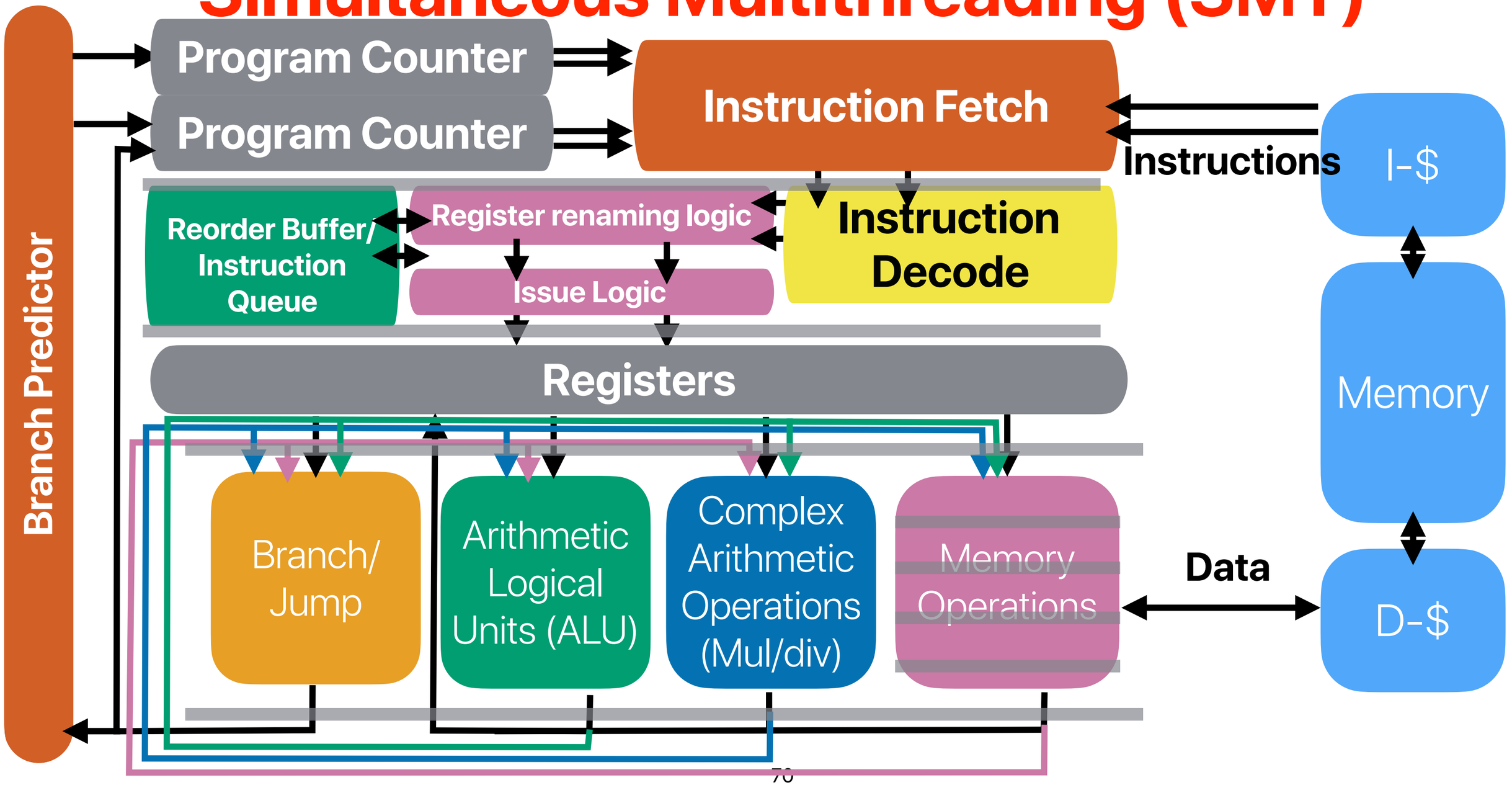
How do we support two running programs in one pipeline?

- We need two program counters
- We need two sets of architectural to physical register mappings
- We do not need
  - Duplicated cache — virtually indexed, physically tagged cache already addressed that
  - Duplicated pipeline functional units — isn’t sharing the whole purpose?
  - Duplicated reorder buffer — you simply need to tag which process the instruction belongs to

Recap: Register renaming



Simultaneous Multithreading (SMT)



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Pros/Cons of SMT

- How many of the following statements about SMT is/are correct?
  - ① SMT makes processors with deep pipelines more tolerable to mis-predicted branches
  - ② SMT can improve the throughput of a single-threaded application
  - ③ SMT processors can better utilize hardware during cache misses comparing with superscalar processors with the same issue width
  - ④ SMT processors can have higher cache miss rates comparing with superscalar processors with the same cache sizes when executing the same set of applications.

  A. 0
  B. 1
  C. 2
  D. 3
  E. 4

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Pros/Cons of SMT

- How many of the following statements about SMT is/are correct?
  - ① SMT makes processors with deep pipelines more tolerable to mis-predicted branches
  - ② SMT can ~~improve~~ the throughput of a single-threaded application
  - ③ SMT processors can better utilize hardware during cache misses comparing with superscalar processors with the same issue width
  - ④ SMT processors can have higher cache miss rates comparing with superscalar processors with the same cache sizes when executing the same set of applications.

  A. 0
  B. 1
  C. 2
  D. 3
  E. 4

We can execute from other threads/contexts instead of the current one

hurt, b/c you are sharing resource with other threads.

We can execute from other threads/ contexts instead of the current one

b/c we're sharing the cache

```
Architecture:                    x86_64
CPU op-mode(s):                  32-bit, 64-bit
Byte Order:                      Little Endian
Address sizes:                   39 bits physical, 48 bits virtual
CPU(s):                          8
On-line CPU(s) list:             0-7
Thread(s) per core:              2
Core(s) per socket:              4
Socket(s):                       1
NUMA node(s):                    1
Vendor ID:                       GenuineIntel
CPU family:                      6
Model:                           151
Model name:                      12th Gen Intel(R) Core(TM) i3-12100F
Stepping:                        5
CPU MHz:                         3300.000
CPU max MHz:                     5500.0000
CPU min MHz:                     800.0000
BogoMIPS:                        6604.80
Virtualization:                  VT-x
L1d cache:                       192 KiB
L1i cache:                       128 KiB
```

SMT in real practice

- Intel HyperThreading (supports up to two threads per core)
  - Intel Pentium 4, Intel Atom, All Intel Core i7s and Core i9s
- AMD RyZen (Zen microarchitecture)
- If you see a processor with "threads" more than "cores", that must because of SMT!

Takeaways: parallel architectures

- SMT processors can better utilize the pipeline resources by allowing simultaneous execution of multiple threads
  - Improved execution throughput
  - May hurt the latency of each thread since we share functional units & cache

Announcements

- Assignment #4 — due this upcoming Sunday
- Very last reading quiz due next Tuesday
- Check your latest grade on https://www.escalab.org/my_grades
  - Please make sure you've refreshed your browser and see the timestamp of the source data to be 8/27

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